

Shining light on the dark matter of pertussis: evidence for an asymptomatic carriage state from a longitudinal cohort of mother/infant dyads

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eLife Assessment

This **important** study provides evidence for asymptomatic *Bordetella pertussis* carriage among mothers in a longitudinal cohort in Zambia, significantly advancing understanding of transmission dynamics. The evidence presented is **convincing**, with strengths including routine sampling irrespective of symptoms and rigorous qPCR methodology, although confirmatory diagnostics would further strengthen the claims. Overall, the study represents an influential contribution to the field of infectious disease epidemiology.

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Abstract

Pertussis remains an enigma partly due to the uncertain impact of asymptomatic cases on transmission. Contributing to this knowledge gap is a lack of high-quality disease surveillance, particularly in those low- and middle-income countries that experience high disease burdens. Here we present analyses based on our prospective longitudinal surveillance of a rolling cohort of 1,315 mothers and their newborn infants in Lusaka, Zambia across 2015 (8,704 unique study visits). We detail the timing, duration, and intensity of qPCR-based IS481 signals in individual subjects and within mother/infant dyads. We find that IS481 signal strength: A) in mothers predicts contemporaneous and future IS481 detections in infants, B) in infants predicts, to a lesser extent, detections in mothers, and C) predicts respiratory symptoms in infants but not mothers. We profile a subgroup of 50 infants and 54 mothers who displayed evidence of persistent colonization (median duration 8 weeks) wherein most mothers were entirely asymptomatic. We also include a critical assessment of qPCR test reliability across IS481 signal strengths. Our results indicate that pertussis transmission routinely occurs between minimally symptomatic mothers and their newborn infants, and demonstrate the routine occurrence of long-duration, mild and minimally symptomatic pertussis infections.

Introduction

The resurgence of *Bordetella pertussis* in a number of countries with high routine immunization coverage (1) has called into question key assumptions about the transmission, immunology, and evolution of this highly contagious respiratory pathogen (2, 3, 4, 5, 1). Chief among these questions is the epidemiological importance of mild and asymptomatic pertussis infections (6, 7, 8, 9, 10). Indeed, despite widespread recognition that mild pertussis is common (2), especially amongst adults (11, 12), its contribution to overall disease transmission and persistence in populations remains poorly characterized (13, 6, 14).

Mild and asymptomatic pertussis has sometimes been considered the “dark matter” of pertussis transmission, owing to its elusive yet potentially significant impacts (9, 15). Prior to modern molecular surveillance, reliable detection of mild pertussis infections was challenging (16, 17, 18) and, in the pre-vaccine era, the epidemiological role of mild pertussis was largely ignored (19, 20, 21). As vaccination rates rose worldwide across the 20th Century, pertussis morbidity and mortality dropped sharply (22, 23, 24, 4). Yet local pertussis elimination has rarely been achieved, even in highly-vaccinated populations, unlike many similar childhood diseases such as measles, mumps, and rubella. Consequently, understanding the determinants of pertussis transmission and persistence remains a pressing global health concern and an area of active public health research (25).

In recent decades, novel molecular surveillance tools have yielded remarkable insights into global pertussis dynamics, from phylodynamics (3, 26) to pathogen evolution (27, 28, 29, 30). Of particular note, qPCR-based testing can detect mild pertussis much more reliably than previous culture-based methods (31, 32, 33). Deployed at scale, qPCR offers the potential for a deeper understanding of when and where mild pertussis is most likely to occur, and how these infections contribute to local and regional disease dynamics. At present, however, high-income countries are responsible for the bulk of routine molecular pertussis surveillance (15). As these countries typically also experience the lowest burdens of most infectious diseases, such concentrated surveillance means that the best-studied populations may offer modest (and hardest-won) epidemiological insights.

Our 2015 Southern Africa Mother/Infant Pertussis study (SAMIPS) sought to bridge this gap through the use of targeted, high-resolution pertussis surveillance in a low-resource urban community (34, 35). Conducted in Lusaka, Zambia across one calendar year, SAMIPS followed a cohort of mothers and their young infants across these infants’ first 3-4 months of life. Each mother/infant dyad attended regularly-scheduled visits at a local public health clinic (PHC), where self-reported respiratory symptoms were recorded and nasopharyngeal (NP) samples were collected. At these visits, routine vaccinations and well-child care were also administered and recorded. Overall, our partnership with the local PHC and establishment of an on-site molecular laboratory enabled us to (I) conduct prospective longitudinal surveillance of the SAMIPS cohort and (II) link detailed clinical records with qPCR testing of nasopharyngeal (NP) samples. Critically, we designed SAMIPS to minimize selection biases by sampling on a fixed schedule rather than driven by apparent symptoms. This framework allowed us to assess the burden both of symptomatic pertussis, as is generally done when only testing children who present with a clinical syndrome typical of pertussis, but also of asymptomatic infections. By sampling mothers and infants on the same schedule, we also gained insights into the epidemiology and clinical presentation of pertussis within mother/infant dyads.

Our previous work in this cohort has profiled a small number of the most sick infants (34), inspected the timeliness of DTP vaccine uptake in cohort infants (35), and described antibiotics administration across the cohort (36). Additionally, we previously quantified evidence for

pertussis infection (EFI) across the full cohort, where we observed a strong correspondence of EFI between mothers and their infants (10 [↗](#)). To our surprise, we also found a strong correspondence between EFI and mild symptoms (cough and/or coryza alone) in infants as well as mothers, suggesting the common occurrence of mild pertussis in infants. We were also intrigued by the detection of apparently asymptomatic pertussis among a subset of infants with strong EFI (10 [↗](#)).

Our previous findings underscored the epidemiological importance of mild and asymptomatic pertussis, especially in low-resource settings, and highlighted the prospect of weak IS481 qPCR signals as sensitive indicators of pertussis infections in both populations and individuals. Our previous work, however, left several key questions unaddressed. Here we explore the detailed time-course of qPCR tests and respiratory symptoms within individual subjects and within mother/infant dyads. We also critically examine the fidelity of our qPCR-based pertussis testing in order to assess the reliability and robustness of our results. Finally, we profile a group of subjects with evidence of persistent colonization. Altogether, our results offer granular insights into otherwise invisible pertussis dynamics in this at-risk population of mothers and their infants.

Results

Study Overview

SAMIPS was designed to enroll, across one calendar year, all newborns and their mothers in Chawama compound, a low-resource peri-urban community with a population of approximately 150,000 people at the time of the study (2015). The study was conducted at the single no-cost public health clinic available to Chawama residents, where subjects were observed at enrollment and (ideally) at six scheduled study visits across infants' first 100 days of life (plus *ad hoc* sick visits). At each visit, we recorded vital signs, antibiotics prescriptions, routine vaccinations, and self-reported and observed respiratory symptoms. Critically, we also collected nasopharyngeal (NP) samples at each visit that were retrospectively tested for pertussis (and other common respiratory pathogens) via qPCR at our lab located at Lusaka's University Teaching Hospital (UTH). For additional details, see Gill et al. (34 [↗](#)), Gunning et al. (35 [↗](#)) and Gill et al. (10 [↗](#)).

Ultimately, SAMIPS enrolled nearly 2,000 mother/infant dyads and conducted qPCR tests on over 19,000 NP samples. We focus here on the subset of 1,315 dyads who attended at least 4 study visits, for a total of 17,408 unique NP samples across 8,704 dyad visits (including 410 unscheduled visits).

qPCR: detection versus signal strength

For clinical pertussis diagnosis, IS481 $Ct < 35$ is a frequently used threshold (15 [↗](#)), yet experimental variation between individual tests, plates, and labs precludes a direct estimate of sample quantity from Ct value in the absence of rigorous controls (37 [↗](#), 38 [↗](#), 39 [↗](#)). Accordingly, we focus here on disease surveillance and epidemiological inference via relative quantification and detection of IS481. In this context, we first classify IS481 qPCR tests as either detections (i.e., $Ct < 45$) or non-detections (N.D.). As before (10 [↗](#)), we further categorize detections by signal strength into three arbitrary groups: weaker ($Ct \geq 43$), intermediate ($43 > Ct \geq 40$), and stronger ($40 > Ct$). *A priori*, we expect that stronger signals are more reliably detected and are more predictive of symptomatic illness, carriage, and disease transmission. We proceed with the assumption that all IS481 detections are *potentially* informative, and directly interrogate the information content of qPCR tests by their signal strength.

Overall, we observe detections in 9% of NP samples ($N=1,540$). Most of these detections consist of weaker IS481 signals (median $Ct = 43.2$); $Ct < 40$ in 10.2% of detections ($N=159$), and $Ct < 35$ for 1.7% ($N=26$). In **Figure 1A** [↗](#), we show the frequency of IS481 detections by Ct value for mothers vs infants (see also **Table 1A** [↗](#)). Notable, stronger IS481 signals ($Ct < 40$) are observed much more frequently in infants (10%) vs mothers (3.7%). We also summarize the strongest observed IS481

signal by subject and dyad (**Figure 1B**), again highlighting a long tail of stronger IS481 signals in infants. In **Figure 1C**, we show the number of mothers, infants, and dyads with one or more detections across their course of study participation. In addition, **Table 1B** tabulates the frequency of co-detection within a dyad's visit (OR=4.9, 95%CI: 4.1-5.9). In all cases, the observed overlap between infants and their mothers is much higher than expected by random chance, illustrating how detections tend to cluster within dyads.

Study Timeline

We employed a rolling enrollment of the study cohort such that each dyad's participation spanned a portion of the full study period. **Figure 2A** illustrates the study participation of six example dyads, highlighting the sometimes uneven timeline of study visits relative to the intended schedule along with infrequent unscheduled visits. This figure also highlights the potential clumping of detections over subjects' study participation, a point to which we return later. **Figure 2B** shows the number of NP samples (and associated qPCR tests) by calendar week and IS481 signal strength, highlighting the changing cohort size across the study period. **Figure 2C** shows the frequency of NP samples by IS481 signal strength from a generalized additive model (GAM) that accounts for the variable cohort size. As previously described (10), we observe a burst of stronger IS481 signals in June 2015 that is preceded by a more gradual surge of weaker signals in the previous six weeks.

Experimental re-testing

During the course of qPCR testing, an opportunistic subset of 1,468 NP samples was tested a second time, including 399 initial detections and 1,069 initial N.D.s. In this subset we observed a very low probability of follow-up detection given initial N.D. of 2.3%, suggesting a low false positive rate. By contrast, in samples where initial qPCR results (Ct_0) were stronger ($Ct_0 < 40$), we found a high probability of confirmation of 86%. In total, this subset revealed a relative risk of confirmation of 21.9 (95% CI: 14.7-32.6, see **Table 2**), indicating very high overall repeatability of IS481 detection. These results prompted us to conduct a systematic evaluation of qPCR reliability as a function of initial signal strength. To do this, we re-tested a random subset of NP samples, stratified by initial qPCR strength across mothers and infants, drawing NP samples per strata across 6 groups of Ct_0 (129 total retested NP samples). As expected, we found no differences between mothers and infants. In **Figure 3A**, we show the re-test confirmation frequency by Ct_0 strata: here, the proportion of confirmed detections rose rapidly from approximately 25% for $Ct_0 > 42$ to 58% for $41 \leq Ct_0 < 40$ and 89% for $Ct_0 < 40$. We also illustrate the overall detection frequency for the full library of qPCR results (8.8%, dashed line), which we henceforth employ as a baseline "null frequency" of detection, representing a conservative upper bound on qPCR false positive rate.

In **Figure 3B**, we consider Ct_0 as a continuous predictor of qPCR re-test confirmation. We use a GAM to predict the odds of confirmation, and then normalize the resulting prediction by the null odds of detection (above) to obtain an estimated mean odds ratio of confirmation (ORoC). Notably, the ORoC's 95% CI ceases to overlap with 1 just below $Ct = 44$, showing that the odds of confirmation exceeds the null odds. At $Ct_0 = 43$, the estimated ORoC increases to 3.1 (95% CI: 1.8-5.5) and at $Ct_0 = 40$ to 19.7 (95% CI: 10.1-38.3). In summary, confirmation was more likely than expected by chance alone (i.e., than the null frequency of detection observed in the full NP sample library) across the full spectrum of CT values, including even weaker signals.

We note that the above results are consistent with the expected stochastic variation intrinsic in qPCR. For example, assuming a realistic PCR efficiency of 1.7, we could expect a Ct of 42 given 10 targets per sample (i.e., at the theoretical limit of quantification; see **Fig. 5** in Ruiz-Villalba et al. (39)). We would also expect Poisson sampling variation in per-reaction target number to yield an approximately continuous decrease in re-test confirmation probability with increasing Ct , as observed here (e.g., see Fig. 9 in Ruiz-Villalba et al. (39)).

Table 1. a) Percent of samples by IS481 signal strength across 8,704 dyad visits. b) Odds ratio of co-detection (OR +95% CI; rows show mothers, columns show infants, N=8,704 unique dyad visits).

(a)					(b)			
	N.D.	[45,43]	(43,40]	(40,15]		N.D.	Detect	OR
Mother	90.7	5.5	3.1	0.7	N.D.	7344	520	1.0
Infant	91.6	4.4	2.8	1.1	Detect	597	209	4.9 (4.1-5.9)

Figure 1. Overview of qPCR test results showing A) frequency of IS481 Ct values (by visit), B) frequency of peak IS481 values (by subject and dyad), and C) number of IS481 detections across full study (by dyad).

A) Intermediate and stronger IS481 signals (color) were much more common in infants than mothers (inset numbers show cumulative frequency from left to right; Ct < 35 not shown). B) More infants exhibit stronger peak IS481 values, while dyad members together frequently exhibit weaker peak signals (dashed line). C) Venn diagrams show the frequency of subjects with ≥ 1 (left) to 4 (right) detections (Ct < 45), stratified by infants, mothers, and both dyad members (shaded region).

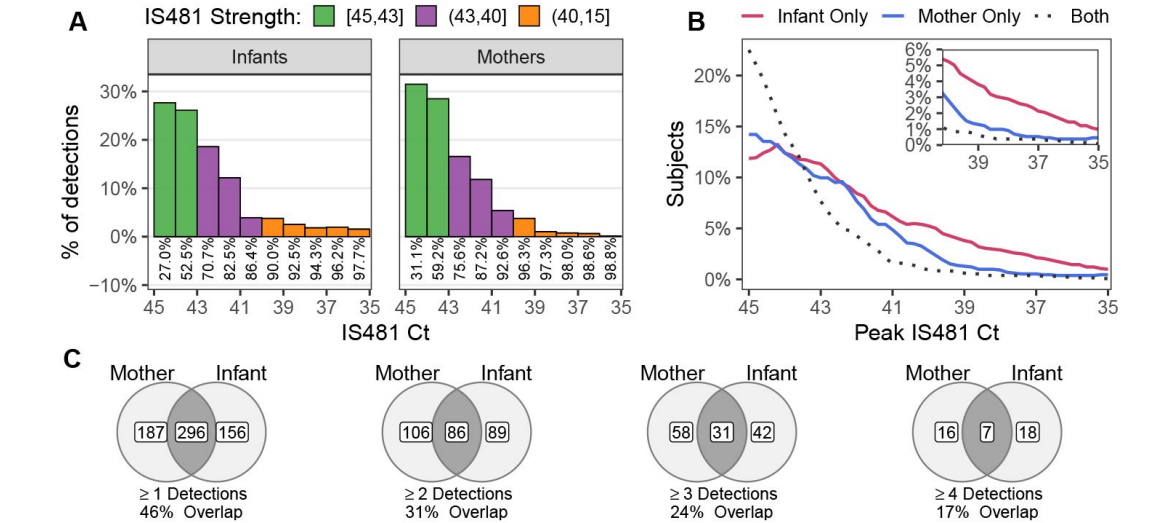


Figure 2.

Study timeline.

A) Timelines of select dyads (mother/infant pairs) showing IS481 qPCR signal strength. Dyad study visits were scheduled at approximately two week intervals (median 16 days). B) Subject visits per week, stratified by IS481 signal strength (color, grey shows N.D.). C) Estimated frequency of IS481 detections by signal strength over time (GAM, mean +95% CI).

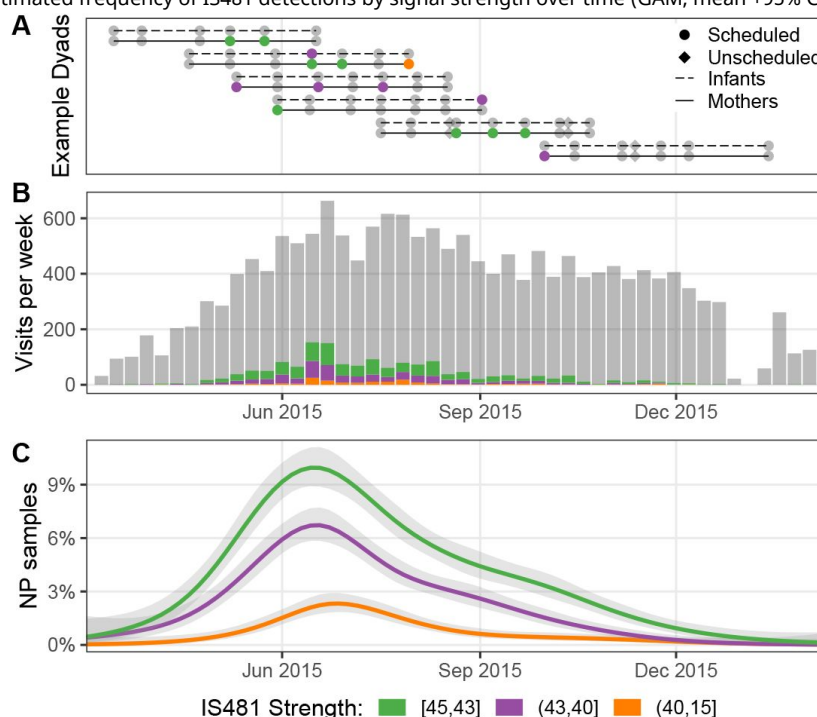


Table 2.

Relative risk of confirmation (columns) given initial detection (rows) in an opportunistic sub-set of 1,468 NP samples that were retested.

	N.D.	Detect	RR
N.D.	1044	25	1.0
Detect	195	204	21.9 (14.7-32.6)

Figure 3.

In a stratified random subsample of 122 NP samples with initial detections, re-test confirmation frequency increased with initial IS481 signal strength.

A) shows confirmation frequency by subsample strata. B) Odds ratio of confirmation by IS481 signal strength (OR relative to null frequency of detection across full sample library; mean and 95% CI estimated from a binomial GLM). Points show initial IS481 Ct stratified by confirmation (shape, color).

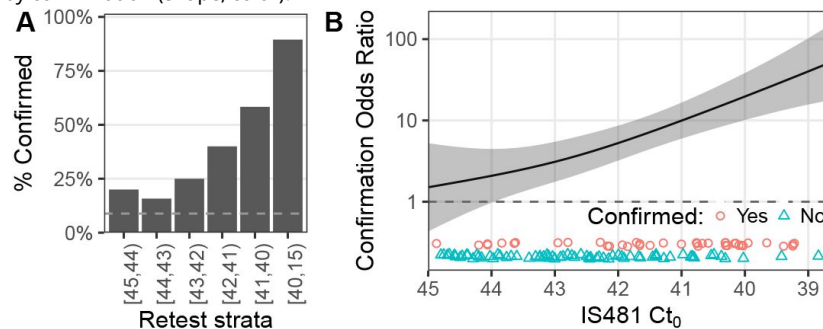
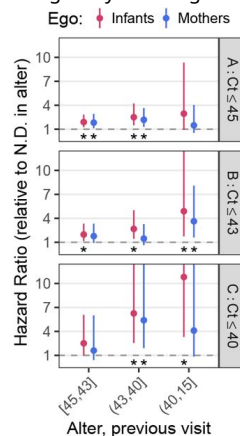


Figure 4.

For each group of events (rows), the mean hazard ratio (HR, relative to the hazard for N.D. in the alter, Y-axis) is shown as a function of alters' preceding IS481 signal strength (X-axis).

Overall, HRs are higher in infants than mothers, and are significantly greater than 1 in most cases (asterisk indicates 95% CI non-overlap with HR=1). Here an event signifies the first occurrence of an IS481 signal at least as strong as indicated (row labels). The infrequent occurrence of stronger IS481 signals yields large CI (bottom row, upper limit not shown).



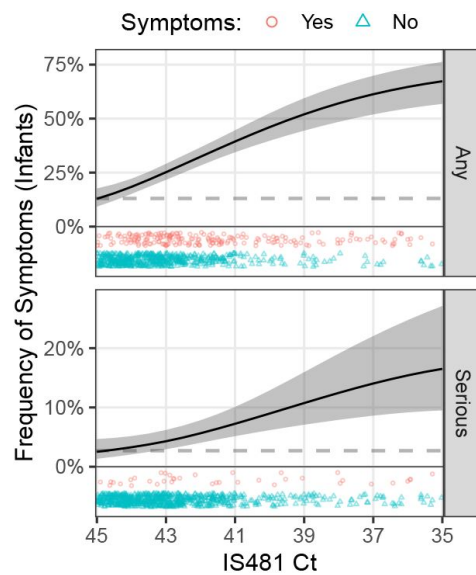


Figure 5.

The frequency of reported respiratory symptoms increases with stronger IS481 signals among infant visits with IS481 detections (N=733), both for any symptoms (top) and serious symptoms (bottom: excludes solely cough and/or coryza).

Curves show mean and 95% CI estimated from GAMs (one per row). Dashed lines shows marginal frequency of symptoms in infants (Any=13%; Serious=2.7%; N=8,705). Points show presence of respective symptoms by Ct value.

Quantifying the information content of IS481 signal strength

The preceding results bolster our confidence that all qPCR tests are *potentially* informative, particularly for epidemiological and public health surveillance where sensitivity often takes precedence over specificity. With this in mind, we next assess the information content of qPCR tests within the longitudinal context of both individuals and mother/infant dyads.

Time to first detection: impact of *alter* on *ego*

We first employ a survival analysis framework to investigate how the time to initial detection in one dyad member (the *ego*) is affected by preceding IS481 signals within their *alter* (i.e., the *ego* and *alter* represents *either* mother and infant, or infant and mother). Here, model covariates include the *ego*'s identity (infant versus mother) and the *alter*'s IS481 signal strength at the preceding visit (N.D., $Ct \geq 43$, $43 > Ct \geq 40$, and $40 > Ct$). Along with the first detection of *any* IS481 signal (event A: $Ct \leq 45$), we also construct survival models predicting the time to first occurrence of intermediate and stronger IS481 signals (event B: $Ct \leq 43$ and event C: $Ct \leq 40$, respectively).

As we move from model A to model C, we focus on greater outcome specificity at the expense of fewer events and lower sensitivity. Overall, we find stronger evidence that infants (*ego*) are affected by mothers (*alter*) than vice versa. **Figure 4** [↗](#) shows the hazard ratio (HR) of each event relative to an N.D. in the *alter*'s preceding visit. Notably, we find elevated HRs in infants even from weaker IS481 signals in mothers' previous visits: event A, HR=1.9 (1.3-2.8 95%CI); event B, HR=2.0 (1.2-3.3 95%CI); event C, HR=2.5 (0.98-6.1 95%CI). And, despite wide confidence intervals from infrequent events, a greatly increased HR of event C in both infants and mothers is predicted from intermediate IS481 signal in their alters: infants, HR=6.3 (2.6-14.7 95%CI); mothers, HR=5.4 (1.9-14.2 95%CI).

These results are broadly consistent with the assumption that mothers more frequently transmit pertussis to their infants than vice versa. Somewhat surprisingly, even weak signals in mothers correspond to elevated HRs in infants. And, while our study design can't exclude the possibility of a shared source of infection, the time-ordered nature of these results offer a potentially useful clinical indicator for infant risk.

Statistical associations within subjects and dyads

While the above survival models suggest a causal role of mothers' IS481 signals on infants and (to a lesser extent) vice versa, these models only consider the first occurrence of each event. We next sought to characterize the full time course of each subject by considering two binary experimental outcomes: 1) future detections **within** subjects, and 2) contemporaneous detections within each dyad visit. We report the relative risk (RR) of each outcome stratified by IS481 signal strength, using N.D. as the baseline group.

For outcome 1, we predict future detections from each subject's *own* IS481 signals at the preceding visit. For infants (**Table 3a** [↗](#)), we find an increase in the RR of detection from 2.3 (95% CI: 1.9-3.1) for weaker IS481 signals ($Ct \geq 43$) to 4.9 (95% CI: 3.7-6.5) for stronger signals ($Ct < 40$), and very little difference between infants and mothers (**Table 3b** [↗](#)).

For outcome 2, we predict detections in the *ego* from contemporaneous IS481 signals in their *alter*. We find a similar pattern as in outcome 1, where IS481 signals of mothers (*alter*) predict detections in infants (*ego*) and vice versa. Yet here we find notable differences between infants and mothers. For infants (**Table 4a** [↗](#)), we again find the RR of detection increasing from 2.7 (95% CI: 2.2-3.3) for weaker *alter* IS481 signals ($Ct \geq 43$) to 8.1 (95% CI: 6.3-10.4) for stronger signals ($Ct < 40$). We observe a more modest impact of infant *Ct* values on detections in mothers (**Table 4b** [↗](#)).

Table 3.

Relative risk of detection (RR +95%CI) per visit is greater given stronger IS481 Ct values in previous visits (no significant difference between (a) mothers and (b) infants).

(a)				(b)			
Infant IS481	N.D.	Detect	RR	Mother IS481	N.D.	Detect	RR
N.D.	6240	525	1.0	N.D.	6159	527	1.0
[45,43]	263	62	2.5 (1.9-3.1)	[45,43]	338	76	2.3 (1.9-2.9)
(43,40]	159	55	3.3 (2.6-4.2)	(43,40]	174	62	3.3 (2.7-4.2)
(40,15]	53	33	4.9 (3.7-6.5)	(40,15]	33	19	4.6 (3.2-6.7)

Table 4.

Relative risk of contemporaneous detection (RR +95%CI) in ego given IS481 signal strength in alter: a) infant risk given mother's IS481; b) mother's risk given infant's IS481.

Overall, stronger signals predict more frequent detections, with modest differences between the predictive strength of IS481 signals in mothers a) versus infants b).

(a)				(b)			
Mother IS481	N.D.	Detect	RR	Infant IS481	N.D.	Detect	RR
N.D.	7344	520	1.0	N.D.	7344	597	1.0
[45,43]	392	85	2.7 (2.2-3.3)	[45,43]	293	91	3.2 (2.6-3.8)
(43,40]	177	92	5.2 (4.3-6.2)	(43,40]	166	79	4.3 (3.5-5.2)
(40,15]	28	32	8.1 (6.3-10.4)	(40,15]	61	39	5.2 (4.0-6.7)

Respiratory symptoms

Our final analysis of the full cohort investigates IS481 signal strength as a predictor of attendant respiratory symptoms. As above, we report RR for mothers and infants separately for *any* respiratory symptoms (predominantly cough and/or coryza). For infants, we also employ a set of GAMs to estimate the frequency of symptoms as a (continuous) response to IS481 signal strength. Here we focus on visits with IS481 detections (i.e., excluding N.D.s) and construct separate models for a) *any* respiratory symptoms and b) serious symptoms (i.e., excluding cough and/or coryza).

Unsurprisingly, we recorded respiratory symptoms far more frequently in infants than in mothers, regardless of IS481 signal strength (**Table 5**). Overall, we find that IS481 signals strongly predict symptoms in infants but only weakly in mothers. For infants (**Table 5a**), we see an increase in the RR of symptoms from 1.5 (95% CI: 1.2-1.9) for weaker IS481 signals ($Ct \geq 43$) to 5.1 (95% CI: 4.3-6.1) for stronger signals ($40 > Ct$). For mothers, we observe a (marginally significant) RR of symptoms that increases from 1.6 (95% CI: 1.1-2.3) for weaker IS481 signals to 2.4 (95% CI: 1.1-5.1) for stronger signals.

Focusing on infant visits with detections, **Figure 5** shows the quantitative response of symptom frequency to IS481 Ct . The frequency of *any* symptom increases from 13% (95% CI: 9-18%) to 46% (95% CI: 40-52%) for Ct values of 45 and 40, respectively. While serious symptoms are much less common overall, again we find an increase from 2.5% (95% CI: 1.3-4.7%, $Ct = 45$) to 9% (95% CI: 6-13%, $Ct = 40$). We note that comparable models for mothers show no statistically significant association between IS481 Ct and frequency of respiratory symptoms.

Taken together, these results quantify how IS481 signals cluster in time, within dyad visits, and correspond with respiratory symptoms. Notably, even the weakest IS481 signals ($Ct \geq 43$) are associated with statistically significant increases in the above outcomes (i.e., RR relative to N.D., **Tables 3-5**). We also observe noteworthy differences between infants and mothers, particularly for the time to first detection (**Figure 4**) and respiratory symptoms (**Table 5** and **Figure 5**).

Persistent colonization: evidence from a subgroup of subjects

The above analyses suggest an important role of false negatives in interpreting the time-course of individual subjects. Yet this potential ambiguity of qPCR signals complicates subject-level interpretation of infection status over time. Throughout the preceding analyses we have examined the full cohort, where IS481 signals are predominantly mild and infrequent (e.g., **Fig 1C**). To conclude, we examine those subjects with strong and/or frequent qPCR signals. In this way we focus on the time-course of subjects with *strong evidence of infection* (10), thereby reducing the relative impact of false negative qPCR tests.

We begin with a detailed view of 20 subjects (10 infants and 10 mothers) with the strongest *peak* IS481 signals, shown in **Figure 6** (subjects ordered by number of detections). Clustering of detections around strong IS481 signals ($Ct < 35$) is evident here, consistent with our previous findings in this cohort of stepwise transitions of IS481 signals within subjects (i.e., **Figure 6** in Gill and Gunning et al. (10)). However, uninterrupted runs of detections are uncommon. We also note that respiratory symptoms are common but not consistently observed in this group of subjects.

With the above in mind, we sought to identify subjects with evidence of *persistent colonization*, which we define here as a cluster of ≥ 3 detections that lacks two or more adjacent N.D. (that is, we permit isolated but not consecutive N.D.s). In this way, we seek to balance sensitivity (including subjects with persistent evidence of IS481) with specificity (focusing on subjects where detections

Table 5.

Relative risk of respiratory symptoms by IS481 signal strength for a) infants and b) mothers.

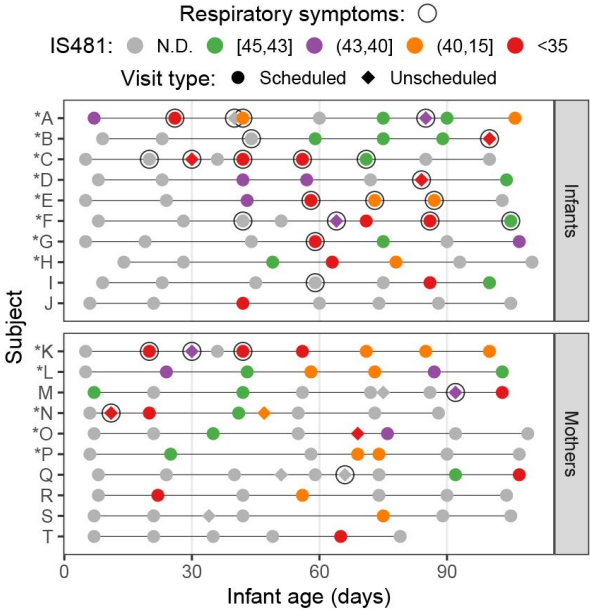
No symptoms: N.S.

(a)				(b)			
	N.S.	Symp	RR		N.S.	Symp	RR
N.D.	7049	923	1.0	N.D.	6412	304	1.0
[45,43]	317	68	1.5 (1.2-1.9)	[45,43]	382	29	1.6 (1.1-2.3)
(43,40]	164	84	2.9 (2.4-3.5)	(43,40]	202	16	1.6 (1.0-2.6)
(40,15]	41	59	5.1 (4.3-6.1)	(40,15]	50	6	2.4 (1.1-5.1)

Figure 6.

Timelines of study visits showing IS481 signals (color) for the ten infants (A-J) and ten mothers (K-T) with the strongest peak IS481 signals within each group.

Peak IS481 Ct in these infants ranges from 18.1 to 32.6 (mean=27.1) and from 30.0 to 36.4 in these mothers (mean=33.7). Clusters of detections are evident that, however, frequently contain N.D.s. Within each panel, subjects are ordered by detection frequency. Open circles mark visits with any reported respiratory symptoms. Asterisks indicate inclusion in the subgroup analysis below (Fig 7), including 8 of 10 infants and 5 of 10 mothers.



are clustered in time). Using these criteria, we identified a sub-group of 50 infants and 54 mothers. Despite not accounting for IS481 signal strength, this subgroup includes 8 infants and 5 mothers from **Figure 6** (marked with asterisks above), showing that our duration-based criteria also capture many subjects of those exhibiting the strongest IS481 signals.

We provide a detailed analysis of this subgroup in **Figure 7**. **Figure 7A** displays the Kaplan-Meier estimated duration of detection, where no difference between infants and mothers is evident (median duration of 55 days, 95% CI: 46-69 (infants) and 50-68 days (mothers)). **Figure 7B** provides a detailed view of censoring, which was especially common in these mothers (where both left and right censoring occurred). Notably, the precision of the above duration estimates are limited by frequent right censoring (i.e., detection at last visit), which was observed in 20 infants and 20 mothers in the subgroup. Regardless, we find substantial evidence of persistent colonization, including 5 infants and 7 mothers with detections lasting ≥ 10 weeks (with right censoring in 5 of each).

In **Figures 7C-D**, we illustrate peak IS481 *Ct* and frequency of respiratory symptoms. **Figure 7C** shows that peak IS481 *Ct* values are *much* higher in this subgroup than the full cohort (i.e., **Figure 1B**), and higher here in infants than mothers. **Figure 7D** highlights the relative sparsity of respiratory symptoms in this subgroup. We were particularly surprised to find that 22% of infants (N=11) and 65% of mothers (N=35) reported no symptoms across their study participation. In addition, of the 28 infants and 16 mothers with strong IS481 signals (*Ct*<40, **Figure 7C**), only 9 infants (32%) and 1 mother (6%) reported *any* serious respiratory symptoms.

Overall, we find strong evidence of persistent colonization in this subgroup, including strong peak IS481 signals and long durations of detection. In addition, frequent censoring shows that our 100-day follow-up period was not long enough to fully characterize the *maximum* duration of colonization, as evidenced by **Figure 7A** and **7C**. Yet we observe minimal respiratory symptoms among this subgroup, even among infants. In addition, we find that a majority of these mothers reported no symptoms at all.

Discussion

Pertussis remains a leading vaccine-preventable disease that is endemic worldwide, and whose global impacts have been underscored by its dramatic post-pandemic resurgence across a range of countries (40, 41, 42, 43, 1). This is despite generally high national vaccine coverage (44, 1), though considerable variation in vaccine uptake and timing exists worldwide (45, 46, 47, 35, 1). Further complicating this picture are major gaps in pertussis surveillance at the national and sub-national level, particularly in LMICs, as well as rapidly shifting diagnostic methods and standards (48, 49, 50). Meanwhile, mounting evidence of vaccine-driven evolution in *B. pertussis*, along with widespread antibiotic resistance in some regions, calls into question whether current clinical and public health tools will remain effective in the near future (51, 52, 26). And while each country faces its own unique challenges, the widespread transmission and evolution of pertussis remains a public health risk to the world at large.

To successfully address the many public health challenges posed by pertussis, our results indicate that a clearer view of national and sub-national pertussis dynamics is needed, especially in low-resource settings where the burden of disease is highest. In this regard, Lusaka, Zambia is broadly representative of the urbanized Global South, where whole-cell vaccines predominate and prenatal maternal vaccination is cost-prohibitive (53).

In our cohort of mothers and their newborn infants, we rarely observed severe disease but were surprised by the high overall frequency of IS481 detections, including a substantial number of intermediate and stronger IS481 signals (**Figure 1**). We commonly observed contemporaneous

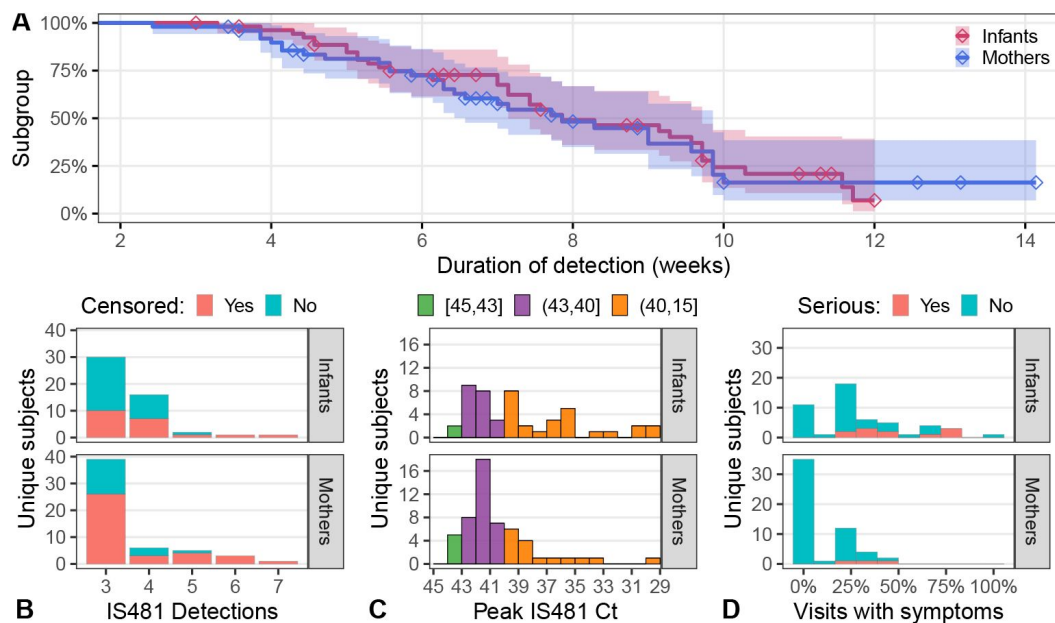


Figure 7.

Subgroup analysis: subjects with evidence of persistent colonization, including 50 infants and 54 mothers.

A) Kaplan-Meier survival analysis of duration of detection (i.e., time interval between the first and last detection) that accounts for right censoring (detection at final study visit, shown by diamonds; includes 20 infants and 20 mothers). Estimated median duration of detection was 55 days. Panels B-D show the number of subjects by (B) number of detections, (C) peak IS481 signal, and (D) frequency of visits with respiratory symptoms. B) Number of detections stratified by censoring (color). For mothers, both left and right censoring are included (i.e., detection at first and last visit). C) Peak IS481 Ct is much higher in infants than mothers and, unlike the full cohort ([Figure 1B](#)), is rarely a weaker signal (green: Ct >43). Not shown: 3 infants with peak Ct ranging from 23.4 to 18.1. D) Most subgroup infants experienced respiratory symptoms infrequently, while most subgroup mothers experienced no respiratory symptoms. Color shows serious symptoms (i.e., excluding simple cough and/or coryza), which occurred most frequently in a small number of infants.

detections within mother-infant pairs (Table 1b), and we documented more frequent respiratory symptoms in those infants (but not mothers) with stronger IS481 signals (Table 5, Figure 5). We could not confidently attribute directionality of infection within dyads. However, our survival analysis points to a greater impact of mothers on their infants than vice versa (Figure 4). Finally, we identified a subgroup of some 50 mothers and infants who displayed evidence of persistent colonization (median duration of approximately eight weeks), in which most mothers were entirely asymptomatic (Figure 7D). Taken together, our results support a picture of episodic but widespread pertussis transmission within the broader Chawama community, and of transmission from mostly asymptomatic mothers to their young infants.

Test reliability, disease surveillance, and public health

Considerable debate exists among pertussis clinical epidemiologists about the use of specific detection thresholds for interpreting IS481 C_t values (54, 55, 56, 15, 50). Our retesting results (Table 2 and Figure 3) emphasize the overall high fidelity of the IS481 qPCR assays conducted in our lab while also highlighting the sampling variation intrinsic in this qPCR assay (57, 55). In modern laboratory settings, this uncertainty is commonly addressed via repeat testing and qPCR efficiency correction (37, 39, 38). However, disease surveillance efforts are often time- and resource-constrained, especially in low-resource settings such as Lusaka. With this in mind, our central goal was *not* to directly quantify IS481 copy number, but rather to reliably identify and characterize pertussis infections in subjects. In this context, our limited qPCR retesting provides a benchmark for repeatability in this system and points to an important C_t threshold ($C_t = 40$) below which stochastic non-detections were rare in our lab.

In assessing the information content of weaker qPCR signals, some caveats are warranted. First and foremost, caution should be exercised in comparing C_t values between labs in the absence of objective standards and/or qPCR efficiency corrections (38, 58). Notably, we suspect that our average qPCR efficiency was lower than that of more well-resourced laboratories, and lower efficiency could partly explain the abundance of very weak signals in our testing. Unfortunately, we did not retain PCR efficiency information as part of our laboratory pipeline. We recommend that future efforts prioritize reporting this information in accordance with modern best practices for Minimum Information for Publication of Quantitative Real-Time PCR Experiments (MIQE) guidelines (37).

We also note that interpretation of weak IS481 qPCR signals must account for the high copy number of this repetitive element. For example, our laboratory protocols specified that 1/50 of the DNA extracted from an NP sample was included in each qPCR test. Given a conservative estimate of 240 IS481 copies per genome, an NP sample that captured 10 bacteria would yield approximately 48 copies per qPCR test, well above the so-called qPCR limit of quantification of ten targets per reaction (see Fig. 8 in Ruiz-Villalba et al. (39)).

The above estimates stand in sharp contrast to the widely-cited claims of Tatti et al. (59), who write: “On the basis of our analytical sensitivity data ... IS481 C_t values in the range of $35 < C_t < 40$ indicate the presence of less than 1 bacterium per reaction, which we consider uninterpretable.” As we have detailed above, C_t values that represent fewer than one genome equivalent contain valuable information and should be expected in any carefully-conducted surveillance effort. We recognize that such weak and potentially ambiguous signals may not be appropriate for clinical diagnosis. However, our results demonstrate that they nonetheless contain valuable information about pathogen presence and infection intensity that can (and should) be leveraged for disease surveillance.

Assessing population immunity (and susceptibility)

An important outstanding question from our study is the role of repeated pertussis infections in immune boosting. If we assume that 2015 was a typical year in Lusaka and that every mother who experienced ≥ 3 detections experienced a clinical infection (**Figure 1C**), then we estimate an annual attack rate of 6.8% of mothers. This corresponds to a mean return frequency of one infection per 14.8 years, which is much shorter than the presumed duration of immunity from natural infection or the whole-cell pertussis vaccination used in Zambia (60, 61). We note that pregnancy impacts immune functioning (62), and that the postpartum mothers in our cohort are not necessarily representative of the broader Chawama population. Nonetheless, our results point to episodically high community prevalence that could, in turn, drive immune boosting this population.

Unfortunately, the absence of a serological correlate of protection (63, 64) makes the confident assessment of population immunity via serosurveys challenging in the best of circumstances. Moreover, the proposition of widespread serological surveillance efforts in low-resource settings faces substantial logistical challenges. From our own field experiences, we found it much easier to increase the cohort size and frequency of clinic visits in the absence of blood draws and processing. With this in mind, we suggest that qPCR-based prospective disease surveillance offers a powerful and cost-efficient tool for revealing heterogeneity in pertussis dynamics across space and time.

Implications for global public health

Pertussis vaccine uptake remains a key country-level metric in tracking infectious disease prevention worldwide, led by the WHO and GAVI (65, 47, 66). However, much less is known about the true burden of pertussis infections in LMICs (53, 67). In this context, pertussis surveillance in Lusaka is broadly representative of urban low-resource settings worldwide, where active surveillance is largely absent and passive surveillance only captures the most severe symptomatic illness. Indeed, the majority of active pertussis surveillance (and research) at present is focused on high-income countries where incidence is least common. In order to fully understand the “pertussis enigma” of sustained disease transmission in the face of high levels of presumed immunity from infection and/or vaccination, we expect that researchers will need to alter existing strategies and broaden their horizons.

As a populous, middle-income and primarily urban country, Zambia offers an evocative example of pertussis surveillance, where no cases have appeared in official WHO reports since 2009 (68). Yet by following a modest-sized cohort of 1,315 mothers and their newborn infants across a single year, we identified 55 infants (4.2%) and 50 mothers (3.8%) with evidence of persistent colonization, most of whom were minimally symptomatic. These results point to a chronically neglected “invisible mass” of minimally symptomatic pertussis incidence below the visible iceberg of officially recorded pertussis cases (69).

We believe our study also offers a template whereby high-quality disease surveillance can be achieved through efficient use of in-country public health resources. At various points we faced challenges in local capacity building and the timely processing of samples. Nonetheless, we believe that our strategy could be implemented at scale in a cost- and resource- efficient manner in a middle-income country, for example, through coordination of public health clinics with a central diagnostic laboratory, as we did. Such an active surveillance program would shed invaluable light on the total burden of pertussis incidence in low-resource settings as well as help characterize the spatial and temporal dynamics that drive sustained transmission. Such a program could also complement local public health efforts, for example, by calibrating ongoing intervention efforts, and could yield further dividends by increasing public trust in government and building local biomedical diagnostic capacity (70).

Materials and Methods

Study Design

We provide a detailed account of study methods, including sample size considerations and inclusion criteria, in Gill et al. (34) and Gill et al. (10). We offer a brief overview here.

SAMIPS was designed as a longitudinal cohort study that followed mother/infant dyads across infants' first three months of life. The study was based in Chawama compound, a densely populated low-resource peri-urban community near central Lusaka, Zambia. SAMIPS sought to enroll all healthy live births that occurred between March and December 2015 in Chawama, and included a public outreach campaign towards pregnant Chawama residents. Subjects were enrolled at the Chawama Primary Health Clinic (PHC), which is the only government-supported clinic in this community and primary source of medical care for Chawama residents. Mother/infant dyads were recruited at their first scheduled postpartum well-child visit (approximately 1 week of age). SAMIPS incentivized mothers to join and remain in the cohort by A) provided concierge (no-wait) routine and acute medical care for participants while enrolled, B) providing mothers with a per-visit travel stipend valued at approximately 7 US dollars and C), providing a small gift of baby supplies at the final scheduled study visit.

Each subject was scheduled to attend seven routine study visits, included a baseline enrollment visit and six follow-up visits spaced at 2–3 week intervals through 14 weeks old (maximum, 18 weeks). Additional unscheduled clinic visits could be initiated by mothers for acute medical care and/or routine well-child care. At each visit, we A) obtained nasopharyngeal (NP) swab samples from both mother and infant, B) recorded self-reported respiratory symptoms a standardized reporting sheet, and C) record administration of infant vaccinations and antibiotics (if any). To avoid possible contamination, routine childhood vaccinations were provided in a separate area of the clinic compound after NP sampling was conducted. We used Unique barcodes from pre-printed sticker books to track study records, thereby linking subjects, clinic visit records, and NP samples. qPCR results were unavailable during the study and thus could not affect clinical treatment.

The SAMIPS study received ethical approval from the Boston University School of Medicine IRB and the ERES Converge IRB in Lusaka. The data sets used in this analysis went through a de-identification process at the request of NIH, and no personally identifiable information is used here.

Sample Collection and Processing

We collected NP samples using flocked-tipped nylon swabs (Copan Diagnostics, Merrieta, California). NP samples were collected daily from the PHC clinic daily and transported to our PCR laboratory at the University Teaching Hospital (UTH) and were stored at -80°C. DNA was extracted using the NucliSENS EasyMag system (bioMérieux, Marcy l'Etoile, France). Extracted DNA from each NP sample was tested for the IS481 insertion sequence and the constitutively expressed human RNase P (RNP) to assess sample collection, storage, DNA extraction, and qPCR reaction. Each 96-well qPCR plate contained a negative and IS481-positive control, along with approximately 46 samples (one each of IS481 and RNP). Reactions were run for 45 cycles on either a ABI 7500 thermocycler (ThermoFisher Scientific Inc, Waltham, MA) or a QuantStudio5 thermocycler (ThermoFisher Scientific Inc, Waltham, MA). We inspected samples run in parallel on both machines and found minimal systematic variation between machines. All primers and probes were purchased from Life Sciences Solutions (a subsidiary of ThermoFisher Scientific Inc).

For sample retests, a random subset of NP samples was drawn from the analysis set sample library stratified by IS481: we selected NP samples from 10 mothers and 10 infants across 5 evenly-spaced IS481 Ct intervals from $Ct = 45$ to $Ct = 40$, plus an additional interval of $40 > Ct$.

Statistical Analysis

Similar to our previous work (10), we focus here on dyads where both subjects had ≥ 4 fully-documented study visits. We exclude 6 mothers and 6 infants from our previous analysis set (i.e., **Table 1** in Gill et al. (10)) where sufficient records for the full dyad was not available. All analysis was conducted in R 4.2.2 (71).

As in previous work, we define IS481 *detection* as any detectable Ct value ($Ct > 45$). We further categorized IS481 Ct values into *IS481 signal strength*: “no detection” (N.D.), weaker ($Ct > 43$), intermediate ($43 > Ct > 40$) and stronger ($40 > Ct$). These divisions were based on manual inspection of the empirical distribution of Ct values (**Figure 1A**) and model results from retest data (**Figure 3B**). Of note, we employ categorical IS481 signal strengths in order to compute relative risks for various quantities using N.D. as the reference level.

We also construct a set of linear models that employ IS481 Ct values as continuous predictors. We exclude N.D.s from these models, i.e., conditioning on detection. Generalized additive models (GAM) were constructed using the mgcv package (72). Confidence intervals for all linear models were computed via estimated marginal means using the emmeans package (73).

We constructed a three separate survival models for our (within subject) time-to-event analyses using a Weibull distribution. For each signal strength category, we predict the time (in days) to the first detection of at least that signal strength. Predictors include whether the subject is an infant or a mother and the alter’s categorical IS481 signal strength at the previous visit (and their interaction). We also tested models that included alters’ respiratory symptoms. Survival models and hazard ratios were fit using the flexsurv package (74). Hazard ratio confidence intervals were constructed via bootstrap simulations using 10,000 samples.

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Additional information

Data Availability

Data available at <https://osf.io/wya75/>

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Reviewer #1 (Public review):**Summary:**

The study investigates the role of asymptomatic pertussis carriage in transmission between mothers and their infants, in particular. The authors used a longitudinal cohort study that involved 1,315 mother-infant dyads in Lusaka, Zambia, and they utilized qPCR-based detection of IS481 to track *Bordetella pertussis* transmission over time. Insights from the study suggest that minimally symptomatic or asymptomatic mothers may act as a reservoir for *B. pertussis* transmission in the infants, thus challenging the traditional surveillance methods that focus on symptomatic cases. Additionally, the study also identified a subgroup of persistently colonized individuals where mothers were majorly asymptomatic despite sustained bacterial presence.

The authors aimed to improve comprehension of pertussis transmission dynamics in high-burden low-resource settings, and they advocated for enhanced molecular surveillance strategies to capture full pertussis infection, including those that might have gone undetected.

Strengths:

The strengths are the use of innovative study design, especially the longitudinal approach and routine sampling, rather than symptom-driven testing that minimizes bias in the study. The methodology was also rigorous and transparent by evaluating the IS481 signal strength to classify pertussis detection and conducting retesting to assess qPCR reliability. There were also important epidemiological insights, and the findings challenge the traditional wisdom by suggesting that pertussis transmission may frequently occur outside of symptomatic cases. The findings also showed its relevance to global health and policy by arguing for the incorporation of molecular tools like qPCR for surveillance of pertussis in low-resource settings.

Weaknesses:

These include reliability on qPCR-based detection without additional validation measures like confirmatory culture or serology. There are also potential alternate explanations for transmission patterns observed in the study such as shared environmental exposure or household transmission. Additionally, there is limited generalizability as the study was done in a single urban site in Zambia. There is also a lack of functional immune data.

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Reviewer #2 (Public review):**Summary:**

In this paper, the authors describe the results of a longitudinal study of pertussis infection in mother/infant dyads in Lusaka, Zambia. Unlike many past studies, the authors assessed the infection status of individuals independently of whether they were symptomatic for a respiratory infection. As a result, this work represents one of the first studies specifically designed to assess asymptomatic transmission of pertussis. Using qPCR, the authors find strong evidence for the role of asymptomatic transmission from mothers to infants and also evidence for long-term bacterial carriage. This work represents an important contribution to our understanding of the global burden of pertussis. Also, it highlights the still under-appreciated role of asymptomatic transmission across many infectious diseases (including vaccine-preventable ones).

Strengths:

Unlike many past studies, the authors assessed the infection status of individuals independently of whether they were symptomatic for a respiratory infection. As a result, this work represents one of the first studies specifically designed to assess asymptomatic transmission of pertussis. Using qPCR, the authors find strong evidence for the role of asymptomatic transmission from mothers to infants and also evidence for long-term bacterial carriage.

Weaknesses:

While I am quite enthusiastic about the work, I am concerned that a number of likely relevant confounders were not discussed and that the broader implications of their findings were not well grounded in the existing literature. For example, I could not find information on the vaccination status of the mothers in the study. Given the conclusions about asymptomatic transmission and the durability of immunity, it is important to know the vaccination status of the mothers. Moreover, did the authors have other metadata on the mother/infant dyads, e.g., household size, vaccination status of household members, etc.? Given the potential implications of more widespread asymptomatic transmission associated with pertussis infection, I believe the authors should better couch their results in the context of the broader debate around asymptomatic transmission.

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